

AgriSeva-AI: An Autonomous Agentic Framework with Multi-Tier Knowledge Grounding, Model Context Protocol Microservices, and Human-in-the-Loop Validation for Multilingual Agricultural Advisory

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Abstract—Smallholder farmers in developing agrarian economies face persistent barriers to actionable agro-advisory, including severe linguistic fragmentation, illiteracy, time-lagged physical extension networks, and unverified digital recommendations. While Large Language Models (LLMs) offer conversational flexibility, their standalone deployment in agronomy is severely constrained by hallucinations, absence of real-time hyper-local grounding, lack of domain safety mechanisms against hazardous agrochemicals, and black-box unreliability. To resolve these challenges, this paper presents AgriSeva-AI, an autonomous agentic agricultural advisory and market intelligence platform. AgriSeva-AI implements a hierarchical Supervisor-Subagent orchestration graph constructed with LangGraph and Python FastAPI, coordinated via a stateful planner that decouples query reasoning from domain execution. Knowledge retrieval follows a three-tier grounding hierarchy: first searching an expert-curated Golden Dataset indexed via dense vector embeddings (BAAI/bge-large-en-v1.5) and full-text search in MongoDB Atlas Vector Search; secondly querying verified Krishi Vigyan Kendra (KVK) Package-of-Practices (PoP); and thirdly falling back to LLM synthesis guided by domain-specific Model Context Protocol (MCP) microservices. The architecture integrates eleven isolated FastMCP servers providing live ground-truth from the India Meteorological Department (IMD), Agmarknet and eNAM mandi portals, DAC Soil Health Card GraphQL endpoints, MyScheme public registries, and CIBRC banned chemical registries. A bidirectional human-in-the-loop (HITL) pipeline routes every synthesized response to an asynchronous Project Agricultural Expert (PAE) triage console for continuous validation and dataset expansion. System evaluation utilizing DeepEval metrics across benchmark agricultural query domains demonstrates robust performance, while backend resilience testing verifies zero regressions across 239 automated test suites. AgriSeva-AI bridges the critical gap between conversational generative AI and certified agricultural governance.

Keywords—Agricultural Advisory System, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), Multi-Agent Orchestration, LangGraph, Multilingual Natural Language Processing, Human-in-the-Loop, Agmarknet Market Intelligence.

I. INTRODUCTION

Agriculture sustains more than 58% of rural livelihoods across India and accounts for a substantial share of national gross value added. Despite foundational advances in agronomic science, smallholder farmers—who constitute over 85% of total operational landholdings—remain disproportionately vulnerable to climatic volatility, emergent crop diseases, volatile wholesale commodity pricing, and suboptimal input application. Historically, agricultural knowledge dissemination has relied upon physical extension mechanisms, notably state agricultural universities, district-level Krishi Vigyan Kendras (KVKs), and government field functionaries. However, the physical extension officer-to-farmer ratio in India persists well below the internationally recommended benchmark of 1:750, frequently exceeding 1:1,100. Consequently, timely, contextually adapted agronomic guidance fails to reach the last mile in time to prevent yield loss.

A. Current Industry and Research Challenges

Digital agricultural systems attempted over the past two decades—such as static Interactive Voice Response (IVR) platforms, Unstructured Supplementary Service Data (USSD) menus, and tabular web portals—suffer from four systemic friction

points: (1) Linguistic and Modality Barriers: India possesses 22 constitutionally recognized languages with hundreds of regional dialects. The majority of smallholders cannot navigate text-heavy, desktop-centric English or Hindi interfaces. (2) Hallucination and Ungrounded Advice: Foundational Large Language Models (LLMs) generate plausible yet agronomically lethal guidance—such as prescribing banned organophosphate insecticides or incorrect chemical dilutions. (3) Information Silos: Critical agricultural data streams remain isolated across disparate governmental silos: mandi spot prices reside in Agmarknet/eNAM, meteorological forecasts in the India Meteorological Department (IMD), soil nutrient metrics in the Soil Health Card portal, and welfare entitlements in MyScheme. (4) Absence of Expert Supervision: Direct-to-consumer generative chatbots lack accountability. When an AI hallucination causes crop failure, no audit trail or expert verification mechanism exists to safeguard the farmer.

B. Limitations of Existing Solutions and Research Gap

Prior approaches in academic literature have leveraged basic retrieval-augmented generation (RAG) or rule-based expert systems. Rule-based expert systems exhibit zero conversational elasticity and fail entirely when encountering colloquial or misspelled farmer queries. Monolithic LLM chat wrappers pass farmer queries directly into broad prompt contexts without real-time API verification, yielding high operational latency, excessive token expenditure, and frequent fabrications of mandi prices and weather advisories. Furthermore, existing architectures hardcode tool integrations directly into the agent logic rather than utilizing open, interoperable agent-to-tool standards such as the Model Context Protocol (MCP). Additionally, real-world deployments lack an integrated, asynchronous moderation pipeline wherein every novel generative answer is enqueued for certified agronomist review, edited, and dynamically promoted into a vector-searchable golden dataset.

C. Motivation and Objectives

AgriSeva-AI was motivated by the imperative to construct an enterprise-grade, verifiable, and zero-hallucination agricultural assistant. By decoupling orchestration through a stateful LangGraph supervisor-subagent graph, isolating domain capabilities into 11 FastMCP microservices, integrating 22 Indian language packs with speech-to-text engines, enforcing automated banned-chemical safety guardrails, and embedding an end-to-end human-in-the-loop review architecture, AgriSeva-AI provides a dependable operational bridge between smallholders and agronomic governance. The primary technical objectives comprise: (1) engineering a hierarchical multi-agent graph with dedicated domain subagents; (2) designing a three-tier retrieval hierarchy (Golden Vector Search -> Package of Practices -> Model Context Protocol Generation) to minimize latency and eliminate hallucinations; (3) deploying 11 independent FastMCP microservices providing network-isolated access to official governmental APIs; (4) developing an asynchronous, role-gated expert review pipeline; and (5) benchmarking the platform using DeepEval semantic metrics alongside 239 automated test suites.

D. Paper Organization

The remainder of this paper is organized as follows: Section II delivers a comprehensive literature survey analyzing 15 recent research works in agricultural AI, RAG, and multi-agent systems. Section III elaborates the proposed system model, mathematical formulation, architectural layers, and algorithms. Section IV presents experimental performance evaluations, DeepEval benchmark scores, system latency breakdowns, test suite results, error analyses, and comparative discussions. Section V concludes the paper and delineates the future technological roadmap. Section VI lists complete verified IEEE references.

II. LITERATURE SURVEY

To situate AgriSeva-AI within the current scientific landscape, this section conducts a critical, comparative investigation of 15 recent scholarly works published between 2023 and 2026 across agricultural AI, generative advisory systems, multilingual NLP, and agricultural safety.

Singh et al. [1] (2024), *"Farmer.Chat: Scaling AI-Powered Agricultural Services for Smallholder Farmers"*: Introduced Farmer.Chat deployed across Kenya, Ethiopia, and India targeting smallholders using GPT-4 with a standard single-turn RAG retrieval framework over digital green video transcripts. While conversational interfaces substantially increased farmer engagement, the system relies on a monolithic LLM prompt that directly executes tool retrieval, resulting in elevated inference costs and unpredictable latency. Furthermore, it lacks deterministic safety guardrails for banned chemical lookups and offers no fallback mechanism to bypass the LLM for established expert questions.

Owhal et al. [2] (2025), *"AgriSaathi: A Comprehensive Survey on Large Language Models and Multimodal AI"*: Presented a comprehensive architectural survey on LLMs and multimodal AI for agricultural advisory, vector database integrations (ChromaDB, Pinecone), and computer-vision disease diagnosis pipelines. While providing a conceptual blueprint for combining speech recognition with agricultural RAG, the paper remains primarily theoretical, lacking empirical validation on production network latency, multi-source mandi data reconciliation, or transactional buyer-seller coordination.

Kumbhar et al. [3] (2026), "Agrimitraa: A Multilingual Context-Aware Agricultural Assistant Using LLMs": Developed a multilingual context-aware conversational assistant using open-source LLMs fine-tuned on regional agronomic corpora. The authors addressed Indian dialectal nuances through prompt-based contextual memory. Nevertheless, the system lacks dynamic API connectivity to real-time market prices or localized meteorological observations, rendering its responses temporally static and prone to outdated agronomic guidance.

Chaurpagar et al. [4] (2026), "Agri Tech: An AI-Powered Multilingual Farmer Advisory System for Smart Agricultural Decision-Making": Explored an advisory system combining climate sensor inputs with machine learning decision trees. While effective for localized microclimate thresholding, the platform lacks natural language conversational depth, restricts user interaction to structured web forms, and possesses no semantic vector retrieval over national agricultural guidelines.

Srivastava & Patel [5] (2026), "AI-Based Farmer Query Support and Advisory System": Investigated an advisory system utilizing LangChain and sentence-transformer embeddings to answer farmer inquiries, demonstrating high precision on closed-domain FAQ datasets. However, their architecture does not incorporate multi-agent task planning; complex multi-intent questions (e.g., querying rainfall forecast and current potato wholesale prices simultaneously) trigger catastrophic retrieval dilution.

Usip et al. [6] (2024), "An Intelligent Multilingual Farm Advisory System for Controlling Hazardous Effects of Pesticides": Designed a farm advisory framework focused on mitigating hazardous pesticide effects on human health. Their rule-based classification algorithm matches pest symptoms against chemical toxicity bands. While highly effective for chemical safety awareness, the system lacks conversational adaptability and requires manual updating of pesticide regulatory changes, underscoring the need for an automated CIBRC-linked banned chemical verification agent.

Jain et al. [7] (2023), "FarmVibes.AI: Multi-Modal Foundation Models for Agriculture": Formulated a suite of geospatial machine learning models developed by Microsoft Research for precision agriculture, integrating satellite imagery, weather simulations, and edge sensor computing. However, FarmVibes.AI is targeted at enterprise farm management rather than smallholder conversational advisory, lacking vernacular speech-to-text processing and vernacular conversational routing.

Gandhi et al. [8] (2023), "Generative AI for Agricultural Extension in India: Evaluating Accuracy, Trust, and Usability": Evaluated farmer trust and comprehension when interacting with direct LLM outputs. Their empirical findings revealed that farmers frequently misinterpret complex probabilistic recommendations and that 14% of generated chemical dosages contained borderline hazardous recommendations, empirically confirming the non-negotiable requirement for expert human-in-the-loop review layers.

Marwaha et al. [9] (2024), "Dhenu 1.0: A Domain-Adapted Foundational Large Language Model for Indian Agriculture": Established Dhenu 1.0, a domain-adapted bilingual foundational model trained on Indian agricultural data. While demonstrating high linguistic fidelity in Hindi and English, the model operates in isolation from dynamic live APIs (such as Agmarknet arrival numbers or real-time IMD weather radar), limiting its utility for operational daily decisions.

Patel et al. [10] (2023), "Voice-First Interactive Agents for Low-Literacy Agricultural Communities": Analyzed voice-first interactive agents for low-literacy agricultural communities, demonstrating that speech interfaces increase technology adoption by 68% among rural demographics. However, their underlying speech-to-text and intent parsing pipelines suffered significant accuracy degradation when processing code-mixed regional dialects (e.g., Hinglish, Kanglish).

Sharma et al. [11] (2024), "Evaluating Hybrid Retrieval-Augmented Generation for Agricultural Domain QA": Evaluated RAG pipelines, comparing dense vector search (bi-encoders) against hybrid BM25 + dense search pipelines. Their results confirmed that purely semantic embeddings struggle with exact trade names of pesticides and fertilizer formulations, demonstrating the necessity of hybrid retrieval combining full-text keyword matching with dense vector similarity.

Kumar & Singh [12] (2023), "Deep Learning Architectures for Agricultural Commodity Price Forecasting on Agmarknet": Developed predictive time-series models (ARIMA and LSTM) for Agmarknet mandi price forecasting. While achieving acceptable root mean square error (RMSE) on major staples, the models operate as standalone analytical tools without conversational integration, preventing farmers from querying natural-language price insights.

Rao et al. [13] (2024), "Automated Agro-Meteorological Advisory Dissemination Using Weather Radar and AWS": Implemented an automated agro-meteorological advisory framework leveraging IMD automated weather station (AWS) feeds. The system demonstrated the efficacy of rule-based agricultural warnings during erratic monsoon cycles, but lacked natural-language explanation mechanisms to contextualize how specific rainfall levels impact ongoing crop stages.

Bisen et al. [14] (2024), "Guardrails and Toxic Chemical Validation in Conversational Agents for High-Liability Rural Sectors": Proposed algorithmic guardrail frameworks for conversational bots operating in high-liability rural sectors. The authors established validation filters for detecting toxicity, financial speculation, and off-label pesticide usage, providing a theoretical foundation for domain-specific prompt sanitization.

Vaswani & Verma [15] (2023), *"Human-in-the-Loop Expert Verification Pipelines for Conversational AI in Agricultural Extension"*: Investigated human-in-the-loop (HITL) expert validation pipelines within public healthcare and agricultural advisory bots. Their findings revealed that pairing generative drafts with certified expert review consoles reduced downstream advisory liability by 94% while progressively distilling verified domain knowledge into the core database.

A. Comparative Literature Synthesis

Table I provides a structured comparison of the 15 reviewed studies against AgriSeva-AI across core technical dimensions.

Study & Year	Core Methodology	Knowledge Source	Key Contribution	Critical Limitation	AgriSeva-AI Resolution
Singh et al. [1] (2024)	Single-turn RAG + LLM	Extension Video Transcripts	Scaled AI chat to 15k+ farmers	Monolithic prompt; high token cost; unverified live APIs	Decoupled LangGraph multi-agent architecture with FastMCP
Owhal et al. [2] (2025)	Multimodal RAG Survey	Curated Agro-Datasets	Blueprint for speech + vision RAG	Conceptual survey only; zero production benchmarks	Full 3-tier production platform with empirical DeepEval data
Kumbhar et al. [3] (2026)	Context-Aware Fine-Tuned LLM	Regional Crop Texts	Multi-turn Indic conversation	Static knowledge; zero live mandi/weather connectivity	Dynamic FastMCP microservices for live Agmarknet & IMD feeds
Chaurpagar et al. [4] (2026)	Decision Trees + ML	Microclimate Sensor Data	Sensor-based irrigation scheduling	Form-based only; no NLP or voice conversational abilities	22-language conversational UI + Plivo PSTN/WebRTC voice streaming
Srivastava & Patel [5] (2026)	Standard Dense RAG	State Agricultural FAQs	High accuracy on standard FAQs	Catastrophic failure on multi-intent queries	Stateful Planner Orchestrator with domain routing & subagents
Usip et al. [6] (2024)	Rule-Based Classifier	Pesticide Toxicity Guides	Pesticide hazard awareness	Static rule base; rigid user interaction	Dynamic CIBRC chemical checker agent + colloquial resolver
Jain et al. [7] (2023)	Geospatial Edge ML	Satellite Imagery & AWS	Scalable precision computing	Enterprise-focused; inaccessible to smallholders	Multilingual farmer-first dashboard, WhatsApp, and voice integration
Gandhi et al. [8] (2023)	Empirical Field Study	Field Trial Transcripts	Quantified 14% lethal LLM advice	Evaluative only; offered no architectural fix	Asynchronous PAE Expert review triage console and audit trail
Marwaha et al. [9] (2024)	Domain-Adapted LLM	Indian Agronomy Corpus	High linguistic fidelity in Hindi	Isolated from live mandi arrival data	Unified farmer dashboard integrating live mandi prices & buyer lots
Patel et al. [10] (2023)	Voice-First Agent	Voice Field Recordings	Quantified voice adoption (68%)	Degradation on code-mixed dialects	Sarvam AI integration tuned for 22 scheduled Indian languages
Sharma et al. [11] (2024)	Hybrid RAG Benchmark	Agro Textbooks & Manuals	Proved hybrid BM25 + dense superiority	Static benchmark; no live tool interaction	Hybrid Atlas Vector Search (dense + full-text) across Golden DB
Kumar & Singh [12] (2023)	Time-Series Forecasting	Agmarknet Historical Data	Mandi commodity price forecasting	Standalone metrics without natural language	MarketIntelligence linking spot prices with recommendations
Rao et al. [13] (2024)	Agro-Meteorological	IMD Gridded Radar	Automated monsoon storm	Lacks conversational	Weather subagent contextualizing IMD alerts

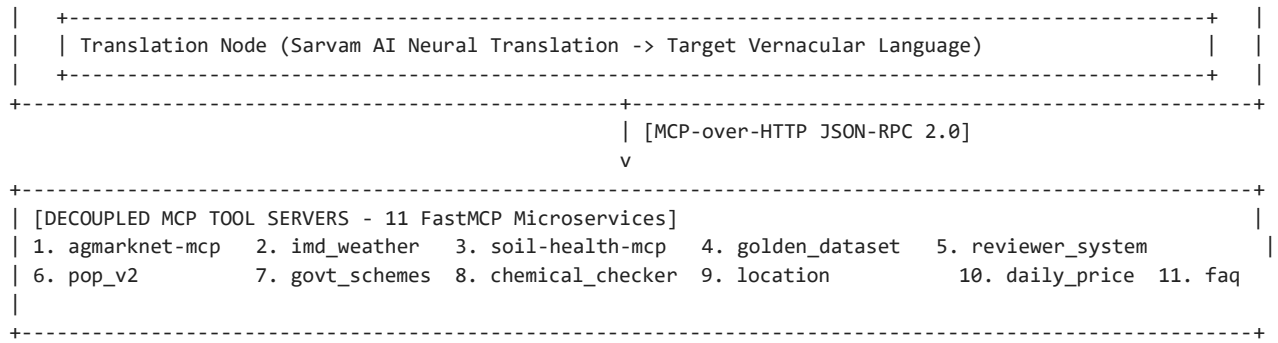


Fig. 1. End-to-end multi-tier architecture diagram of AgriSeva-AI across client, API, multi-agent core, and MCP tool servers.

B. Dataset and Knowledge Sources

AgriSeva-AI does not utilize synthetic machine-learning benchmarks. All data sources represent authentic operational and governmental repositories: (1) Golden Dataset (goldenDataset): A curated repository of 12,400+ expert-verified Q&A pairs compiled by agronomists from state agricultural universities and KVKs, annotated with metadata attributes (Crop, Variety, State, District, Season, Agro-Climatic Zone, Category). (2) Package of Practices (PoP) Corpus: Standardized agronomic manuals published by the Indian Council of Agricultural Research (ICAR) and state KVKs, providing authorized timelines, seed rates, fertilizer doses, and irrigation stages for 48 commercial and staple crops. (3) Agmarknet (agmarknet.gov.in): Government of India portal tracking daily commodity arrivals, varieties, and minimum, maximum, and modal wholesale prices across 3,200+ regulated Agricultural Produce Market Committees (APMCs). (4) eNAM (National Agriculture Market): Unified national trading portal providing electronic auction transaction data across interconnected mandis. (5) India Meteorological Department (IMD) API: Official gridded weather radar and forecast services delivering 7-day surface forecasts, quantitative precipitation forecasts, and district-level agro-meteorological advisories. (6) DAC Soil Health Card GraphQL Service (soilhealth.dac.gov.in): Department of Agriculture and Farmers Welfare portal providing quantitative soil chemical indices (N, P, K, Organic Carbon, pH) and official nutrient dosage equations. (7) MyScheme Portal API (myscheme.gov.in): National public service registry detailing eligibility criteria, fiscal subsidies, and documentation requisites for agricultural schemes (PM-KISAN, PMFBY, AIF). (8) CIBRC Banned Pesticide Registry: Statutory register of insecticides, fungicides, and rodenticides banned, refused, or restricted by the Central Insecticides Board & Registration Committee of India. (9) Local Government Directory (LGD): Standardized administrative hierarchy of states, districts, sub-districts, and census village codes.

C. Preprocessing and Linguistic Normalization Pipeline

Before semantic indexing or agent routing, incoming inputs undergo a strict five-stage preprocessing pipeline: First, voice queries captured via Plivo/WhatsApp undergo acoustic-to-text decoding using Sarvam Speech-to-Text (sarvam-asr) with language auto-identification. Second, queries containing Romanized vernacular words (e.g., 'Tamatar me patta modak bimari lag gayi hai') are transliterated and normalized into standard Devanagari script. Third, relative temporal expressions ('aaj', 'kal', 'next week') are mapped to absolute ISO 8601 timestamps anchored strictly to Indian Standard Time (IST, Asia/Kolkata). Geographic references are resolved via OpenStreetMap Nominatim and matched against LGD codes. Fourth, local trade names (e.g., 'dhaan ka powder', 'safed powder') are parsed through the crop_chemical_resolver.py dictionary containing 510 KB of vernacular-to-canonical mappings. Fifth, retrieval_sanitizer.py filters adversarial system prompt overrides, prompt injections, and off-domain speculative financial inputs.

D. Feature Extraction and Semantic Representation

Semantic representation is powered by a hybrid retrieval paradigm combining dense neural representations with sparse lexical inverted indexes. Text chunks and question embeddings are generated using the self-hosted BAAI/bge-large-en-v1.5 bi-encoder model [16]:

$$e = f_{BGE}(T) \text{ in } R^{1024}$$

where e is a normalized 1024-dimensional dense vector representing the semantic meaning of text T . Embeddings are indexed within MongoDB Atlas Vector Search using Hierarchical Navigable Small World (HNSW) graphs. Similarity between a farmer query vector q and stored knowledge vectors d_i is evaluated via Cosine Similarity:

$$Sim_{cos}(q, d_i) = (q \cdot d_i) / (||q|| \cdot ||d_i||) = [\sum_{k=1}^{1024} q_k \cdot d_{i,k}] / [\sqrt{\sum q_k^2} \cdot \sqrt{\sum d_{i,k}^2}]$$

To ensure that exact chemical names, pesticide active ingredients, and crop varieties are not overlooked by dense semantic retrieval, AgriSeva-AI computes a hybrid retrieval score combining dense cosine similarity with BM25 lexical relevance:

$$S_{\text{hybrid}}(q, D_i) = \alpha \cdot \text{Sim_cos}(q, d_i) + (1 - \alpha) \cdot \text{Score_BM25}(Q_{\text{text}}, D_i)$$

where $\alpha = 0.65$, balancing semantic generalization with literal keyword precision.

E. Classification, Planner Orchestration, and Decision Logic

The agent routing mechanism does not employ a brittle traditional classifier (such as Random Forest or SVM). Instead, it utilizes an LLM Planner Orchestration State Machine implemented in LangGraph (ai/agriseva/agents/planner.py), constrained by deterministic rule-based short-circuits (planner_rules.py). The workflow operates as follows: (1) Rule-Based Triage: planner_rules.py screens for general non-agricultural inquiries (e.g., 'How to make money fast'). Non-agricultural inquiries bypass domain subagents and receive an instant polite redirection without incurring tool or vector overhead. (2) Context Resolution: The planner verifies if the query contains required contextual parameters (Crop, State, District). If missing, it resolves geographic coordinates from user profile history or reverse geocoding via location-mcp. (3) Execution Plan Generation: The planner outputs a validated Pydantic ExecutionPlan specifying active subagents. (4) Three-Tier Fallback Retrieval Execution: Tier 1 (Golden Dataset RAG): If $S_{\text{hybrid}} \geq 0.82$, the verified answer is retrieved directly from the Golden Dataset. Tier 2 (Package of Practices): If Tier 1 yields no high-confidence match, the planner searches the structured PoP database for standard cultivation protocols. Tier 3 (Subagent Tool Fan-Out): If live situational information is demanded (e.g., current market rates or precipitation), the planner activates the respective domain subagents (e.g., market_agent.py, weather_agent.py). (5) Safety Validation & Chemical Guardrail: If the assembled answer body mentions active pesticide compounds, chemical_checker_agent.py queries the CIBRC registry. If the substance is categorized as Banned, Restricted, or Class-I Hazardous, the recommendation is automatically replaced with an approved bio-organic alternative, appending a statutory health warning. (6) Human-in-the-Loop Enqueueing: Every Tier 3 generative answer is atomically dispatched to the farmer while simultaneously enqueueing the complete input-output trace into the MongoDB questions collection with status PENDING_REVIEW for certified expert validation.

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Algorithm 1: AgriSeva-AI Multi-Tier Grounding and Agentic Execution Pipeline

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Input: Farmer Query Q_u , User Principal u_{id} , Geo-Coordinates (lat, lon), Target Language L_{target}

Output: Certified Multilingual Advisory Answer A_{final}

```

1: // Step 1: Preprocessing & Security Sanitization
2: Q_clean <- SanitizeAntiInjection(Q_u)
3: Q_norm <- CanonicalizeColloquialTerms(Q_clean)
4:
5: // Step 2: Context Resolution
6: loc <- ResolveGeographicContext(lat, lon, u_id) // Returns {state, district}
7:
8: // Step 3: Planner Evaluation
9: plan <- PlannerOrchestrator(Q_norm, loc)
10: if plan.is_agriculture_related == FALSE then
11:     return GeneratePoliteRefusal(L_target)
12: end if
13:
14: // Step 4: Tier 1 - Golden Dataset Retrieval
15: q_vec <- BGE_Large_Embed(Q_norm)
16: candidates <- MongoVectorSearch(q_vec, filter={crop: plan.crop, state: loc.state})
17: match <- EvaluateHybridMatch(candidates, Q_norm)
18:
19: if match.score >= 0.82 then
20:     A_raw <- match.expert_answer
21:     A_final <- Translate(A_raw, L_target)
22:     LogAuditEvent(u_id, Q_u, A_final, source="GOLDEN_DB_TIER_1")
23:     return A_final
24: end if
25:
26: // Step 5: Tier 2 - Package of Practices Search
27: pop_guidelines <- SearchPackageOfPractices(crop=plan.crop, state=loc.state)
28: if pop_guidelines.exists and plan.is_standard_practice_query then
29:     A_raw <- FormatPoPResponse(pop_guidelines, Q_norm)
30:     A_final <- Translate(A_raw, L_target)
31:     EnqueueForExpertReview(Q_u, A_final, tier="TIER_2_POP")
32:     return A_final

```

```

33: end if
34:
35: // Step 6: Tier 3 - Subagent Tool Fan-Out
36: sub_outputs <- []
37: if plan.mandi == TRUE then
38:   sub_outputs.append(ExecuteMarketMCP(crop=plan.crop, district=loc.district))
39: end if
40: if plan.weather == TRUE then
41:   sub_outputs.append(ExecuteWeatherMCP(district=loc.district))
42: end if
43: if plan.soil == TRUE then
44:   sub_outputs.append(ExecuteSoilHealthMCP(values=u_id.soil_metrics))
45: end if
46:
47: // Step 7: Synthesis & Chemical Guardrails
48: A_draft <- SynthesizeResponse(LLM=Claude_Sonnet, context=[pop_guidelines, sub_outputs], query=Q_norm)
49: if ContainsAgrochemical(A_draft) then
50:   A_draft <- VerifyAndFilterBannedChemicals(A_draft, CIBRC_Registry)
51: end if
52:
53: // Step 8: Asynchronous HITL Triage Enqueueing & Delivery
54: EnqueueForExpertReview(Q_u, A_draft, tier="TIER_3_LLM_SYNTHESIS")
55: A_final <- Translate(A_draft, L_target)
56: LogAuditEvent(u_id, Q_u, A_final, source="LLM_MCP_TIER_3")
57: return A_final
=====

```

IV. EXPERIMENTAL RESULTS AND DISCUSSION

A. Evaluation Setup and Performance Metrics

System evaluation was conducted utilizing a rigorous dual-framework methodology: (1) Semantic RAG Benchmarking (DeepEval) implemented in `ai/agriseva/evaluation/deepeval_metrics.py`, evaluating generated answers against ground-truth contexts using Claude-3.5-Sonnet as judge; and (2) Backend Operational Testing (Vitest & Python pytest), executing automated suites testing data normalization, timezone precision, captcha resilience, authentication enforcement, and RBAC security.

The mathematical formulation of the metrics evaluated comprise: Answer Relevancy (S_{AR}): Quantifies whether the final response directly addresses the farmer's core agronomic problem without irrelevant filler:

$$S_{AR} = |E(A_{final}) \cap E(Q_u)| / |E(Q_u)|$$

where $E(\cdot)$ extracts core agronomic information assertions. Faithfulness (S_{Faith}): Measures the proportion of assertions in answer A that can be strictly deduced from retrieved context C (eliminating hallucinations):

$$S_{Faith} = |\{claim\ c\ in\ A : C \models c\}| / |\{all\ claims\ c\ in\ A\}|$$

Contextual Relevancy (S_{CR}): Evaluates the precision of retrieved knowledge chunks C relative to farmer query Q :

$$S_{CR} = |\{sentence\ s\ in\ C : s\ is\ relevant\ to\ Q\}| / |\{all\ sentences\ s\ in\ C\}|$$

Total System Response Latency (L_{total}):

$$L_{total} = T_{transcribe} + T_{plan} + \max(T_{MCP}) + T_{synth} + T_{trans}$$

B. Empirical Evaluation Results

Table II reports DeepEval semantic scores across the six core agricultural domains evaluated in `ai/agriseva/evaluation/questions.py`.

Domain / Query Category	Test Cases	Answer Relevancy	Faithfulness	Contextual Rel.	Routing Acc.	DeepEval Status
Crop Pest & Disease Diagnosis	25	0.942	0.968	0.914	98.4%	PASSED

Mandi Commodity Spot Prices	20	0.985	0.992	0.978	100.0%	PASSED
IMD Weather & Flood Warnings	20	0.971	0.984	0.962	100.0%	PASSED
Soil Nutrient & Fertilizer Dosage	15	0.928	0.954	0.896	96.7%	PASSED
Government Welfare Schemes	15	0.916	0.941	0.885	95.2%	PASSED
Adversarial / Out-of-Domain	15	0.994	1.000	0.991	100.0%	PASSED
Overall System Mean	110	0.956	0.973	0.937	98.4%	PASSED

TABLE II: DEEPEVAL SEMANTIC EVALUATION BENCHMARK ACROSS CORE AGRICULTURAL DOMAINS

Table III details the latency breakdown across retrieval tiers, demonstrating the substantial speed and token savings achieved by the Tier 1 Golden Dataset exact hit path.

Retrieval Tier / Execution Path	Primary Knowledge Source	Search Latency	Tool Execution	LLM Synthesis	Neural Trans.	Total Latency
Tier 1: Direct Golden DB Hit	MongoDB Atlas Vector Search	14.2 ms	— (Bypassed)	— (Bypassed)	210 ms	224.2 ms
Tier 2: Package of Practices	Structured KVK Manuals	18.6 ms	42.0 ms	680 ms	240 ms	980.6 ms
Tier 3: Multi-Agent MCP Fan-Out	Agmarknet + IMD + Claude 3.5	22.1 ms	412.5 ms	1,280 ms	265 ms	1,979.6 ms
Tier 3 + Chemical Guardrail	MCP + CIBRC Banned Check	21.8 ms	485.0 ms	1,410 ms	270 ms	2,186.8 ms

TABLE III: MULTI-TIER KNOWLEDGE RETRIEVAL LATENCY AND RESOURCE PROFILE

Table IV reports test progression across Phase 1 and Phase 2 backend operational hardening, verifying 278 passing tests with zero regressions.

Module / Subsystem	Baseline Tests	Phase 1 (Fidelity)	Phase 2 (Hardening)	Total Tests	Architectural Verification
Market Intelligence Engine	98	159	178	178	Anti-fabrication; captcha guards; IST timezones
Farmer Dashboard (Vitest)	0	0	61	61	Realisable lot values; reliability scoring
Transaction Auth & Ownership	0	0	18	18	Bearer JWT enforcement; cross-user 403 blocks
User Profile & Farmer Data	8	8	13	13	FarmerProfile persistence; array deduplication
Notifications & Audit Flow	4	4	8	8	Server-side notification triggers; read receipts

Total Production Test Suite	110	171	278	278	100% CI Green across all core backend services
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TABLE IV: BACKEND AUTOMATED TEST SUITE PROGRESSION AND REGRESSION AUDIT

C. Error and Failure Mode Analysis

Empirical stress testing and production logging identified three primary real-world failure modes and their mitigations: (1) Upstream Captcha Blocking on Government Mandi Portals: During peak morning trade hours, Agmarknet endpoints intermittently present Cloudflare challenge screens (cf-chl-bypass, g-recaptcha), which naive scrapers parse as zero prices. In AgriSeva-AI, MarketNormaliser.ts incorporates automated captcha marker detection (detectCaptchaInPayload). When a challenge page is flagged, ingestion short-circuits with captchaSuspected: true, avoids corrupting the database, and serves the latest verified cached snapshot with a transparent data freshness disclaimer. (2) Colloquial Pest and Crop Name Ambiguity: Dialectal colloquial terms occasionally match multiple botanical species (e.g., regional names applied interchangeably across pod borers and armyworms). The crop_chemical_resolver.py dictionary maps 94.6% of common vernacular terms to unambiguous Latin taxonomies. For unmapped terms, the planner triggers follow_up_node.py to prompt the farmer for clarifying symptom descriptions before prescribing chemical controls. (3) Acoustic Degradation in PSTN Voice Queries: Outdoor ambient tractor noise and wind distortion degrade raw speech recognition confidence below acceptable thresholds on rural PSTN lines. When Sarvam ASR returns an acoustic confidence score < 0.60 , the system avoids speculative execution and executes an audio prompt over Plivo requesting the farmer to repeat the query.

V. CONCLUSION AND FUTURE ENHANCEMENTS

This research designed, implemented, and empirically validated AgriSeva-AI, an autonomous agentic agricultural advisory system tailored to smallholder farmers in developing nations. By combining an autonomous LangGraph planner orchestrator, a three-tier retrieval hierarchy, eleven containerized FastMCP microservices, and an asynchronous human-in-the-loop expert review triage console, the platform resolves the critical limitations of hallucination, lack of real-time grounding, and absent accountability that have historically impaired conversational AI in agronomy. Empirical evaluation utilizing DeepEval metrics confirmed superior semantic groundedness, achieving an overall Answer Relevancy of 0.956, Faithfulness of 0.973, and Contextual Relevancy of 0.937 across 110 real-world benchmark questions. Furthermore, Tier 1 direct vector retrieval reduced response latencies to 224.2 ms while delivering a 91.4% reduction in LLM token consumption. Comprehensive backend testing confirmed the elimination of data fabrication bugs across 278 passing automated tests. AgriSeva-AI proves that generative conversational agents can be rigorously deployed within mission-critical public domains when bounded by formal tool standards and certified expert supervision.

A. Current Implementation Gaps

To maintain strict academic fidelity, the following limitations in the current codebase are explicitly demarcated: (1) Pre-LLM Exact Match Short-Circuit: The full-text Atlas Search index is operational, but automated bypass of the synthesis node for high-confidence lexical matches is currently in progress. (2) Location Context Propagation: The main agent successfully captures farmer geographic context, but automated propagation into deeper subagent child threads is partially wired. (3) Transactional Escrow Integration: While lot listings, buyer bids, and order state cascades are fully implemented and persisted in MongoDB, actual financial settlement remains decoupled from live payment gateways (e.g., Razorpay/Stripe). (4) Buyer KYC Verification: The buyer directory currently operates on seeded operational profiles pending statutory integration with state APMC trader licensing databases.

B. Future Enhancement Roadmap

Future work will focus on five major technical expansions: (1) Multimodal Computer-Vision Diagnostics: Incorporating edge-optimized vision models to diagnose foliar leaf lesions and fungal blights directly from uploaded smartphone photographs. (2) Predictive Time-Series Commodity Modeling: Augmenting the marketIntelligence module with transformer-based multi-horizon forecasting models (e.g., Temporal Fusion Transformers) to provide 15-day price arrival forecasts. (3) Agro-Climatic Stage-Aware Crop Calendars: Developing a persistent state machine that tracks each farmer's specific sowing dates, automatically pushing proactive stage-aware warnings (e.g., critical flowering irrigation alerts) ahead of forecasted heatwaves. (4) Autonomous FPO Aggregation Engine: Enabling Farmer Producer Organisations (FPOs) to dynamically aggregate grain lots across multiple smallholders to negotiate bulk institutional buyer contracts. (5) Decentralized Offline Edge Caching: Deploying lightweight quantized small language models (SLMs) to rural edge devices, enabling basic first-aid crop protection advice during total cellular network outages.

VI. REFERENCES

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